Intro of Show and Guest

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A lot of our problems, they start with the gut microbiome. And food and nutrition is the most powerful lever

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that we have that we could pull in order to revitalize and restore that microbiome and get it back to health.

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Because when we do that, these diseases wash away. A person who's born in 1990 is four times more likely to be

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diagnosed with rectal cancer and two times more likely to be diagnosed with colon cancer during their lifetime

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compared to their parents if they were born in 1950. No way. 95% of Americans, they are

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wildly deficient in fiber. The benefit of consuming fiber is that you get a healthier microbiome that's producing

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short- chain fatty acids. If we want an effective immune system, we need a strong microbiome. The food choices that

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you make today will start to impact your microbiome by tomorrow. So many mental disorders do in fact

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originate in the gut. I can point you to the studies that show that if you look within the state of

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these mood disorders, the gut microbiome is disturbed. there's damage to the gut microbiome which we call dispiosis.

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If I'm listening to this podcast and yes, I'm concerned about my gut health and yes, I'm concerned about my microbiome, where does somebody start

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and what does a healthy microbiome diet look like? The people with the healthiest guts.

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Ultimate human. Hey guys, welcome back to the Ultimate

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Human Podcast. I'm your host, human biologist Gary Brea, where we go down the road of everything anti-aging,

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longevity, biohacking, and everything in between. And I am super super pumped for today's guest. Um, as you heard in the

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in the bio, he's a very, very accomplished physician, moved from the dark side to the bright side now. And,

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uh, we just had we did a pre-podcast in my kitchen before we jumped on the podcast, and my staff kept saying, "Hey, you guys need to stop talking and get on

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the podcast so everybody can hear this." Um but uh we are going to go down the road um of the gut the gut microbiome of

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some of the links to all kinds of conditions that you might not think are related to actually having a healthy gut

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microbiome. So I am super super excited to welcome Dr. Will be well Dr. Will

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Bullowitz um to the podcast which is why they talk call him Dr. Willby. Welcome to the podcast brother. Yeah, thank you,

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Gary. It's a pleasure to be here, man. Um, I am such a fan uh of your work. I went deep down the rabbit hole of

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looking into some of your work and and even watching some of the other podcasts that you've been on. So many fascinating

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topics about microbiomes and cancer tumors and and and the link between the

Dr. Will B's Transformative Journey

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gut microbiome and all kinds of conditions, some of which I've professed for a while about mental conditions that

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actually can originate from the gut. And so there's so many places that we can

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go. I I want to just kind of start with the basics. You know, I I I always say

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that I feel like the most influential people that I have on my podcast are

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people that were on their way in life to doing something else and there was an event, something in their life, light

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bulb went off, there was a tipping point, they met someone, they had a profound experience, they had a painful

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experience, an addiction they conquered, something happened in their life and they had this tada moment and and they

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went down a a new path and It seems like you know you h you you know being

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traditionally trained as a medical physician. Um but now your viewpoint

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seems to be so much more holistic, so much more dynamic, so much more about the whole organism. Can you just talk a

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little bit about your journey, how you got to where you are? Um, I think it's interesting that you say that because I

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think that profound things need to happen in our lives in order to inspire us to work towards greater things. And

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at my lowest moment, I was actually being tremendously blessed. So this was

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going back uh about 12 years ago like roughly 2012 2013. So to contextualize

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this, I spent 16 years of formal education to become a board-certified gastronurologist.

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And um during that time I was working 100 hours a week, you know, and uh had

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to give everything that I had to my career. That's what they asked of me. And I had nothing left. And of what I

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did have left, I said, "You know what? I just want to feel good, even if it's for five minutes." Wow.

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And so I ate a junk food diet because it worked so perfectly in my life. Yeah.

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So I I had no money as a gastroenterenterologist. He had a junk food diet. Oh, 100%. In fact, most gastronologists

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do. I mean, but Gary, the number one cause of death among cardiologists is heart disease. Wow.

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Yeah, it's true. I did not know that. So, but in this moment there, I was in my early 30s. I had no money, so a fast

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food diet was inexpensive. I had no time. Well, I could get food real quick. Um, I enjoy food. I liked the way that

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it tastes. Like, I I won't deny that, right? And so, it fit into my life so

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perfectly. It required no effort. Mh. And that's what I lived on. And it was like twice a day. And it was kind of

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funny because here I am to talk about the power of a healthy diet. Yeah. And I was the guy among my circle of friends.

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Yeah. That if they were to say who's the least likely out of all of us to be one day sitting in the chair with Gary Brea.

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Yeah. It's you. Yeah. So are you in a hospital setting, a clinic setting? Like I was in a hospital. So I I did my uh

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medical school at Georgetown. I was at Northwestern for my residency and then the University of North Carolina for my GI fellowship. So, I was working in

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these hospitals taking care of really sick people and and what ended up happening is that

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um my health got away from me. So, I was a high school athlete. I still, you know, thought of myself as

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being an athlete, but it kind of feels like I woke up in my early 30s and I like took a look in the mirror one day

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and I didn't recognize the person in front of me. Wow. And I was in a dark place. I was 50 pounds overweight. Um I had extreme

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anxiety. I had extremely low self-esteem. um high blood pressure, high

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cholesterol, and I was quite miserable to be totally honest with you. And it was funny because if you looked

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at me on paper, you would think that my life was perfect. Yeah. And if you knew me personally, you would

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know that I was not doing well. Making decent money, respect to being a physician, probably top of your career.

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I won the highest award at Northwestern among 60 doctors. It's so competitive. We have 4,000

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applicants. I was number one. And I won the top award at UNCC, too. On paper, it looked

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great. I was in a dark place. And and so in that moment when I needed help,

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I didn't want what I had been trained to do for 16 years. I didn't want the pill.

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Right. Right. I wanted a root cause solution. And I needed to figure out how to do

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that because I didn't know how to do that. I hadn't been taught how to do that. Right. And I initially tried to exercise

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my way out of this. I'm a very type A person. So I was willing to go and hit the gym for an hour a day.

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And in addition to hitting the gym for an hour a day, also either run a 5K or swim 50 laps in the pool.

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Wow. That did that six days a week. Okay. And I got faster. I could run further. I

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was stronger. But my gut was just as big. Ah. And so clearly I needed more than that.

The New Science of Microbiome

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And ultimately it was changing my diet that allowed me to undergo a radical transformation in my health that was so

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profound that as a medical doctor who cares deeply about the people that I

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work with or work for, I had no choice at that point. This has to be something that I bring to my

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patients. So I became quite obsessive about studying this science that I had discovered which was the new science of

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the microbiome, right? Because that's where I think the root of my issue was. And so and and I brought

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this into my clinic and from 2014 on I was treating patients with irritable bowel syndrome, ulcerative colitis, Crohn's disease, acid reflux, and

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watching them have as Yeah. And they're having as as radical of a life transformation as I had.

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You mean and you were when you say you were treating them had you already started to migrate away from like the

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corticosteroids and imunosuppressants and you were migrating towards the diet side of things to to help treat them. So

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in conventional gastronology the diet is not a part of the plan at all. Right. Right. So conventional gastronology

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astounding to me. I mean the only re reason why we eat is to feed our microbiome. Right. I mean we don't really eat to feed ourselves. We we eat

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to feed the intermediary in our gut and then that feeds us to oversimplify things. But

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well I 100% agree and and it's actually shocking to me how highly intelligent people don't see that.

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Right. Right. Because they will see including a gastroenterenterologist. I mean, talking about the world's leaders who

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are standing at the podium at an inflammatory bowel disease conference and teaching on inflammatory bowel

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disease and they don't believe that diet has anything to do with inflammatory bowel disease. Yeah. You know, you can't read the label

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from inside the jar. You know, it's like sometimes we're so buried and so mired inside of our own

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beliefs that we we can't lead read the label in, you know, in the jar that we're stuck inside of. Somebody said

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that to me the other day. I'm like, it's a great analogy. Um, you know, I I really try not to allow myself to get

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too dogmatic or pigeonhold or religious about any one direction or philosophy

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because um I meet so many amazing people like yourself and you know they bring

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forward evidence and you're like wow that really but but in your community or

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previous community um there's really not a whole lot of room for free thinking. No, that's true. I think there's not a

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lot of room for free thinking and you know and and just to be totally clear this is not I'm not rejecting

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conventional medicine right if I I don't either by the way yeah if I need if I need conventional medicine then I would have it I think

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that the issue that we have is an overreiance on pills and procedures and using that as our first line approach to

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problems right where the root of the problem is not the absence of the pill or the procedure

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right the root of the problem is something that's deeper inside of us right that I would make the argument and I

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think that you would probably support this I think that a lot of our problems particularly in the gastronurology space

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they start with the gut microbiome and and food and nutrition is the most powerful lever that we have that we

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could pull in order to shape and change and revitalize and restore that

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microbiome and get it back to health because when we do that when we successfully get the microbiome back to health these diseases wash away

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right right so from my perspective to me the approach to people who need um uh

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therapy for their health problem should not start with medication. It should start with addressing the root of the

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issue which is diet and lifestyle. Yeah, I totally agree with you and and if we just took a step back for a

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moment. Um let's talk about all of the incredible systems inside the body that

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are touched from the microbiome, right? I mean, first of all, the, you know, I I often remind people that a lot, you

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know, our intestinal tract is technically outside of our body, right? I mean, and and what's what's like our skin, um, but it's sort of folded

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inside, right? You know, down the down the throat and and out the rectum, but

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the barrier separating that external environment from our internal environment is only a single cell layer.

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Yes. Right. And what determines whether or not that layer is being assaulted or

Gut Microbiome's Impact on Our Health

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is benefiting has a lot to do with our microbiome. Yes. Um and it's a very delicate layer. I mean um you know I

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remember when when I first read Dr. Pearl Motor's book years ago was kind of ahead of its time. It was called the gut

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brain axis um and the gut brain connection. And I was really shocked as

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a human biologist that the gut could have so many profound effects on mental

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illness, on autoimmune diseases, on um immune competency. I mean, talk a little

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bit about, you know, the basis for, you know, our gut microbiome and and why

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it's so important. What are the other systems that it's connected to? So, so listeners really tune in and say, I need

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to pay it really pay attention to this. Yeah. Well, so first of all, we evolved with these microbes. So the very first

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human, whoever that was millions of years ago, they had a microbiome and we

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co-evolved in a way where we grew to trust them. And it's very clear to me as a medical doctor that this is true

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because they um have actually inserted themselves into human physiology

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in a way where we require them in order to properly function as humans. So if we

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in theory had were sterile and did not have these microbiomes, did not have a microbiome, we would be in big trouble

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very quickly. So and the gut is this place. So you just talked about how the

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gut is external to our body, which is true. It's weird to think about that. It is. Um, but the gut is also the place where

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we interact with the environment. We interact with the outside world. The skin is a barrier. It's a wall. But the

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choices that we make when we put something into our mouth, chew it and swallow it, that's based upon trust. You

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trust that whatever it is that you put into your mouth is not toxic, right? And in fact going to nourish your body

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and then you send it down and it will come into contact with these gut microbes and inside they're mostly

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concentrated Gary inside of our colon. All right. And so it's at the very end. At the very end. Yeah.

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And there's 38 trillion microbes that exist in that specific location. Wow. Now, 38 trillion is a hard number

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for us to sort of um comprehend. So, let me try to put this into perspective. If we were to on the most clear, like we're

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if we were way up north in Canada on the most clear night and we looked up and we saw all the stars in the sky,

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I would need to take 380 380 galaxies full of stars,

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tighten them up, and insert them into your colon, and that's how many microbes you have living inside of you right now.

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Wow. They outnumber your human cells. You are less than 50% human. They make up 99.5% of your genetic code.

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Your genetic code is less than 1% human. It is 99.5% your microbiome.

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Wow. And they're connected to so much. It starts with their interactions with our

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food. So they help us to unpack our food and they make nutrients available to us

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that otherwise wouldn't be available. So all food comes into contact with these microbes, right? And this is to me like the

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essence of life. Yeah. Is like, you know, you you're not alive unless you get access to nutrients.

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Ever wonder how elite athletes recover faster and stay energized? It's the science of cold water therapy. Hi, I'm

Feeding Your Microbiome

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Gary Brea, and I want to introduce you to Plunge. Plunge helps to soothe your sore muscles, boost your energy levels,

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and improve mental clarity. It's a powerful way to support your immune function, and enhance your mood. Ready

15:02

to transform your health? Take the plunge. Change your life. Visit plunge.com and shop now.

15:08

Now, let's get back to the Ultimate Human podcast. Yeah, we were saying before the podcast like you you don't eat to feed yourself.

Inside the Intestine: Microbes' Existence as an Ecosystem

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You eat to feed your microbiome. That's right. And then they're going to feed you. And then they're going to feed you. You got to be kind to these guys.

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And these nutrients that they create for us um are bioactive and are able to

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actually exert effects, influences throughout our entire body. So, it's more than just within our intestine.

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Now, if we were to zoom in, you know, the microscope and look at the intestine itself, what we would discover is you

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would see, oh, there's these microbes. Mhm. And they're as alive as you and I are, all right? And they have personalities

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and they have clicks and circles of friends and they have unique skill sets. Certain ones do certain things

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and they work in teams and they have food preferences. They don't all like the same food. So,

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guess what? They sound a lot like us. Right. Right. and they're existing there inside of us in an ecosystem

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and surrounding them is the gut barrier. And as you said, it's a single layer of cells. On the other side of that gut barrier is

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70% of your immune system. Yeah. Right. And the reason why your immune system is there is because this is where

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we're interacting with the outside world. So if there's a problem, that's where the action is going to be. That's where the action is, right? So they're and and they're communicating

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to one another. The microbes are talking to the immune system. the microbes are actually maintaining that gut barrier

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that protects the immune system and that protects our bloodstream so that we don't allow toxins or things

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into our blood that shouldn't be there. But then the the these bioactive molecules which we'll talk about some of

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them like short- chain fatty acids for example our microbes create these things. They enter into the bloodstream

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and within seconds they're circulating throughout our entire body, right? And having healing effects including our

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metabolism, including our insulin sensitivity, inflammation and inflammation and our brain and even

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Gary, our brain, our uh blood brain barrier. So, our blood brain barrier is actually conceptually the same as the

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gut barrier in that it's also a single layer of cells with the same tight junctions that

Gut Microbiome's Disturbance Potentially Causing Mental Disorders

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hold the cells together. And so, um, when people eat well, they

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may notice more energy. They may also notice that they're cognitively more focused. And that's the effects of your

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gut and these bioactive molecules, not only on your brain, but on your blood brain barrier. We talked about it in the kitchen before the podcast, you know, um

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that so many mental disorders, I'm just going to sort of group them all together for a minute. Um ADD, ADHD, OCD, manic

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depression, bipolar, anxiety, anxiousness, all of these things that we might be labeling as a mental condition

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or mental um disorder do in fact originate in the gut. meaning the the

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the the neurotransmitters that provide those different moods and emotional states originate in the gut.

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I think there's a highly compelling argument and if we take a look at each of these, every single one of the ones that you just mentioned, I can point you

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to the studies that show that if you if you look within the state of these um mood disorders, the gut microbiome is

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disturbed. There's damage to the gut microbiome, which we call dispiosis. Right? So, and we see that in each of

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these cases. Now, there are some who will make the argument, oh well, um, these are associations,

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right? That that they're true at the same time, but it's not necessarily proof of causality, but I I actually think that like as far

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as we can take it, there's evidence that would suggest that this is in fact legit. Um we have uh

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fecal transplant studies where they will take the fecal transplant of a person who is clinically

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depressed and transplant it into a mouse and then the mouse will change its mood

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and and start to demonstrate behavior consistent with the spectrum of how mice

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manifest depression. Right. Right. Right. And and the opposite I I' I've read from fecal transplant studies as well where it can

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actually they can improve mood and emotional state and even improve some of

Healthy Microbiome Diet and Lifestyle

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these conditions because through fecal transplant by rein inserting a a new

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microbiome. Um, so when we if we were to take a step back for a second and say, um, if I'm

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listening to this podcast and yes, I'm concerned about my gut health and yes, I'm concerned about my microbiome, like

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where does somebody start and do we just go and test it? Um, and then do we

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adjust our diet accordingly? Are there some things I could add to my daily routine that would help ensure that I'm

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putting healthy preo postbiotics into into this system and

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improving its function? Like where where does where does somebody start and what is a what is a healthy

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microbiome diet look like? Yeah. Um so the good news is that as

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powerful as this is, the microbiome is extremely forgiving.

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So, we have research that shows us that the food choices that you make today will start to impact your microbiome by

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tomorrow. Wow. Now, this is not to make the claim that you're completely overhauling your microbiome. That takes time,

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right? That would be reshaping it. But um but what we do know is that those changes will start to take place

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immediately. And if you start to make changes with consistency, consistency is a very important concept within the

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microbiome. Yeah. So like that one meal, you know, or doing like that quick, you know, five

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day thing, that's not going to get the job done. What we need is we need to identify sustainable habits,

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sustainable choices that you're actually willing to do and that will help you to reshape the microbiome.

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And there's multiple ways that we can shape this microbiome. These are things that you talk about and some of them are dietary and we can

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talk about those concepts, prebiotics, things like this, right? So some of them are dietary but some of them also are through fitness

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through exercise right through sleep through aligning with the circa with our circadian rhythm right through uh

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meditation or bre or breath work. So the the the you know the interesting thing

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to me is that the um I enjoy seeing the way that

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people like for example you um address these issues because in all cases you

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say you know um 30 minutes of exercise within 30 minutes of waking up and I say okay I can show you how this affects the

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microbiome and why that makes sense. Yeah that's great. I would love to I would love to link some studies to to to

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that. And I think that there's so much confusion out there. There's so many oriented approaches um out there that um

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you know I I eat a lot of kafir vegetables. I post it on Instagram all the time. I talk about it. I I can tell

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when I'm not eating um fermented foods. Um and you know, when I was a kid, I did

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didn't like it. We were joking around before the podcast about my grandmother used to can everything. And I swear to

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God I would I walked into her basement when I was 10 years old, came back when I was 19 years old and the same can of beets was on, you know, the jarred

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vegetables, but she, you know, fermented um vegetables all the time and it was a

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part of my life growing up. And then I got way away from that in in high school and college. Um, and only recently I've

Power of Fermented Foods

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I've returned to that. And the difference in how I feel and and even just the diameter of my gut when I'm

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regularly eating um, you know, fermented vegetables is is night and day. And I

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see it in my kids too, even in my staff because I I you know, I feed them when they're here. Um, so talk a little bit

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about the the kinds of foods. I think there's a lot of myths surrounding fiber. You know, was is is fiber really

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as important as we um you know, it's it's held out to be the difference between soluble and insoluble fibers.

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What are ways to increase soluble and insoluble fibers in in my diet so I can actually get more of these short- chain

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fatty acids so I can get the internal benefits that I'm really after? Yeah, it really kind of starts with what we're

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putting in the front end. Yeah, it it 100% does. And you know if if we want to touch on you know if I

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were to offer up two main takeaways for the listeners at home let's start with what you just put on the table which is

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fermented foods. Okay. All right. And then we can move on to other stuff like fiber. But um fermented

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foods when I wrote my first book Fiberfield it came out in 2020 and I I had an entire chapter dedicated to the

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power of fermented foods. Ferment fermentation is transformation right? The food starts as one thing and

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it ends as something different. Right? And the transformation is taking place because of the microbes. They're

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using their specialized enzymes that they have to unlock the nutrition, the

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nutrition within the food. Yeah. So sauerkraut is a great example of this where it starts off as as salty cabbage

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and it ends up as this tart, crunchy, acidic, delicious thing that has now

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different B vitamins. B vitamins. Yeah. um unlocking the vitamin C, making the polyphenols more

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bioavailable, unique forms of fiber that we call exopolysaccharides.

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And then of course the part that we all focus on with fermentation, which is the live microbes. So the um the jar of sauerkraut has

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turned into an ecosystem and that ecosystem exists much like the ecosystem inside of us with a balance of

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microbes where the good guys out outweigh the the bad guys. Yeah. And it and I could take you to,

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you know, the store, Gary, and show you the probiotics that people pay a lot of money for that contain the exact same

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microbes that you would find in a jar of sauerkraut. Right. And you're probably more likely to have them be live.

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Oh, 100%. Yeah. In a jar of sour. So, so the so to me this was intuitive.

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Like this makes complete sense that this is a gut health food. And there of course is the tradition, you know, the

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these are celebrated foods from across the planet. Mhm. But it wasn't until much more recently that actually at at Stanford University,

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some of my friends who are there, um, Professor Christopher Gardner, Professor Justin Sonnenberg, his wife Erica Sonnenberg,

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they combined and they did a randomized control trial. And in this randomized control trial, they were measuring people's gut microbiome. They were

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looking at measures of inflammation and they had them dramatically increase their fermented food intake.

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Now, the baseline is basically zero. Um, in America, people either don't consume

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fermented food at all or maybe they do a little bit of kind of junky yogurt. Right. Right. And so in this study, they ramped

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up their fermented food intake. And the exciting part about this was that in just eight weeks

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they were able to increase the diversity within their gut microbiome and they lowered measures of

Cancer Therapies and Microbiome

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inflammation which by the way does not come as a surprise to me because when we increase the diversity in our microbiome. A hearty healthy microbiome

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is a resilient microbiome which results in a resilient healthy immune system that's lower in inflammation.

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Yeah, that is so that is so powerful. You know, I I heard you um on a podcast

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and I was I was listening to you talk about a really fascinating concept that

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um there's now evidence that even um cancer tumors have their own microbiome.

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Yeah. Um and the microbiome may be predictive

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of its susceptibility to chemotherapeutic treatments. That's right. That one blew me away.

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Yeah. Um because you know I always think of the microbiome not just microbes but the microbiome as being just you know

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inside the gut not um inside of something like like a tumor. Um it made

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a lot of sense to me what you said but talk to me a little bit about that research and the promise that that might

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have for you know oncology treatments. We're transitioning to a new phase with

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the microbiome. Mhm. It used to be that the microbiome was this novel thing where you would kind of

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take a look at the test and you'd be like, "Oh, that's interesting." But you didn't really know how to use the information. Yeah. Yeah. Exactly. I mean, I've even

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looked at some of these microbiome tests and I'm like, "Well, I don't know how we repopulate that flora, this flora, that flora. I mean, take a bunch of

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probiotics, but and I think that's what most people think, too." So, well, now we're entering into this

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new phase where um we're actually starting to use this information. or using what we know in terms of

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manipulating the microbiome in order to help people achieve their health goals. And there's no better example of this

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than than within cancer therapy because these of course are desperate people, right? They're like fighting for their

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lives. They're desperately hoping that the treatment they receive from their doctor like chemotherapy or imun imunotherapy,

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which I'll explain in a moment, is going to help them to to um eliminate this

27:46

tumor and and the the carcinogenic cells in their body. Mhm. Um so the new phase is that you know you

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you mentioned uh that we now see that a tumor has a microbiome

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and that increasingly you can predict response to therapy using the information that exists in the tumor

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microbiome. Now that's not quite at um uh that concept is not quite out in

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prime time just yet. Right? This is early days in terms of that idea, right? But if we were to shift our focus to a

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different idea that is actually out in practice now, it's amunotherapy and it started with

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melanoma, but now this is being applied to multiple different forms of cancer. The new way that we approach cancer,

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which by the way is proving to be more effective for most cancers, is rather than directly trying to crush and

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destroy the cancer cells, which is chemotherapy, we are now flipping to a new idea, which

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is why don't we just empower the immune system? Yes. And allow the immune system to do what

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it does, right? Which is eliminate cancer cells. Yeah. It's astounding. I I actually just

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had a client of mine, I'm not licensed practice medicine, so this was not my doing, but a client of mine, um very

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very prominent um woman, 80 years old, was diagnosed with um stage four

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metastatic colon cancer in February. Um and had a DNA mismatch. So the immune

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system had recognized the cancer as a foreign body and was already beginning to create antibodies to it and did

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imunotherapy and in less than seven months was completely cancer-free at 80

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years old imotherapy. And you would have thought that somebody like that would have gone into highdosese radiation,

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chemotherapy, maybe exploratory surgery and not likely at 80 years old to have

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to have survived that. But she is alive right now. Um and it was it was 7 months

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um from the time and I'm not again saying that imunotherapy is the only thing that can do for for a cancer but

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what's really interesting is that you're now talking about the immune system and

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the strength of the immune system and specifically using the microbiome as a way to target some of these therapies

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maybe even decide how treatment resistant a tumor is going to be. So the concept is proof that our immune system

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requires a healthy microbiome in order to be healthy itself. Okay? Right? If we want an effective immune

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system, we need a strong microbiome. And it started um really this is research

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coming out of MD Anderson looking at melanoma. Melanoma is the number one cause of skin cancer death in the United

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States. And they were um treating people with amunotherapy. This idea that you empower

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the immune system to do the work for you. And they noticed that people who

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received antibiotics were not doing as well. H and so now at first they said well maybe

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they're just more sick. So they started to dig into this in more detail and they discovered oh no what's happening here

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is that the antibiotics are whacking out the microbiome and without the microbiome intact the

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imunotherapy doesn't work. Wow. So they they followed this up with a

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study looking at fecal transplant. Mhm. And they said, "Okay, so if antibiotics are bad, then maybe if we can restore

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the microbiome with a fecal transplant, we could get better results." They took a group of people who had failed

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amunotherapy. These are people who are desperate. They're they have no choice at this point. Like they failed the best therapy

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that we have and they gave them a FICO transplant.

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And now this was a a series of patients and they were shocked by the number of

Feeding Microbes with Fiber

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people that ended up responsive to imunotherapy after receiving a fecal transplant after previously failing the

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same treatment. Wow. Okay. So then they they followed this up with an additional study.

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Um and I think this is where the exciting science is show is is emerging for us and it can lead us into our our

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second sort of nutritional conversation. Mhm. The the second study, they took a look

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at what they were eating and they specifically looked at fiber

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intake. Wow. So now fiber, as we'll discuss, is um

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the key to feeding and fueling the gut microbiome. And when you feed it and

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fuel it fiber, the microbes break down the fiber. It stops being fiber and they

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transform it into fat. Short chain fatty acid. short- chain fatty acids, butyrate, acetate, and

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propriionate. And these short- chain fatty acids have effects throughout the entire body, including directly on the

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immune system and repairing and restoring the gut barrier. Wow. Right. So now these are really important

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and powerful things for people who are sick. Um so anyway, they took you need to feed them the right things

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in order for them to produce the short- chain fatty acids, right? I mean, the right types of fiber. you you get the

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the benefit of consuming fiber is that not only do you get a healthier microbiome, but you get a healthier microbiome that's producing short- chain

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fatty acids. And you need the fiber in order to produce the short- chain fatty acids, right? Right. They don't they don't just, you know, operate out of

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nowhere, right? So, um, and what they found in this study, again, getting back to melanoma

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with the amunotherapy, is that people who were consuming more fiber had a profound survival advantage. And for

Risks of Colon Cancer

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every five grams of fiber that they had in their diet, five grams. This is not a lot. That's not a lot at all.

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For every five grams of fiber that they had in their diet, they had a 30% survival advantage. For every five grams of

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every five grams. Wow. That's not the only Wow. That is seeing findings like this.

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There's similar there's similar research to suggest uh this with colon cancer really because you know colon cancer is

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skyrocketing right now. Um, I mean, you had job security in your your career.

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Yeah. Um, did you do surgery as well? Yeah, I I did colonoscopy. I've I'm actually very serious about colon cancer

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because um I'm the guy who's diagnosed many people even at a young age and it's it breaks my heart like these these were my

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patients right at at a young age. So, you're on the prevention bandwagon now. Oh, I'm on the prevention bandwagon, but I'm also

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on a holistic bandwagon of saying, look, this the the the solution is prevention,

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but a prevention strategy is not just colonoscopy. Right. Right. A prevention strategy starts with

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what we put into our mouth. Because this is colon cancer, why is it shifting? Because it's shifting to younger people.

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Mhm. And for those who haven't heard the headlines, a person who's born in 1990

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is four times more likely to be diagnosed with rectal cancer and two times more likely to be diagnosed with

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colon cancer during their lifetime compared to their parents if they were born in 1950. No way.

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So in 40 years time we have four times the risk of rectiom is disturbed in the development of

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colorectile cancer. On the flip side, if you want to prevent colorectal cancer,

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the way that we do this is with short- chain fatty acids. Really, there are multiple mechanisms, multiple

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mechanisms that short- chain fatty acids are actually able to destroy tumor cells.

35:03

Short chain fatty acids are actually able to destroy tumor cells. Destroy tumor cells because we are constantly replicating our bodies. Our

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body makes mistakes in replication. It's normal to create abnormal cells,

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but our body has to have the capacity to get rid of them. We have to be able to identify them and get rid of them.

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And if we don't, if we don't see it, the zombie cells, the scinesscent cells, those cells that are just not

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metabolically healthy. Exactly. And and they start it starts off as one cell. And if our body

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identifies it and eliminates it, then there's no problem. And this is what happens on a daily basis, like as we're sitting here talking to one another.

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Right. Right. But if our body misses it because the immune system is not healthy and

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because we don't have things properly oriented then that one cell becomes two cells becomes four cells becomes eight

35:49

cells and you give it enough time and it turns into a tumor. Wow. Right. So and it starts on a microscopic level like that.

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Mhm. So um short- chain fatty acids have the ability to directly uh destroy tumor

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cells through multiple different mechanisms. And this is important information because we're talking about

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the benefits of short- chain fatty acids on a number of different levels. We get them from fiber,

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right? 95% of Americans are not not just mildly

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deficient in fiber. They are wildly deficient in fiber. Wow. So to put numbers to this, men like like

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you and I, we should be consuming at least 38 grams of fiber per day. The average man in the United States is

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getting 17. All right? Not even half. So, not even half. Yeah.

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The average woman in the United States is getting 15 grams of fiber and she should be getting 25.

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So, about 60%. So, we're nowhere close to where we need to be. And then we ask our question, why

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do we have such a colon cancer problem? And the solution that our government comes up with is things like colonoscopy

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programs, which I'm completely in favor of. We need to protect ourselves. But what they don't talk about at all, which

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makes no sense to me, is that what if we addressed the fiber deficiency that exists in the United States. This is

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more than just colon cancer. This is heart disease, esophageal cancer, breast cancer, strokes, diabetes, Alzheimer's disease.

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We could keep going. Yeah. Type three diabetes. Yeah. Um, so you really think that having a

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high fiber diet is is a non-negotiable? I think that I think that we should

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start by addressing the fiber deficiency, right? So getting 38 grams of fiber is not a high fiber diet,

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right? That's just actually normalizing. And I wonder how many people are actually getting to that that point.

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Getting to 38 grams of fiber. Um among men it's uh I think about 3%.

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3% about 3% get to 38. So and again that's not a high fiber diet. We consider that a high fiber diet, Gary, because we have

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normalized something that's very abnormal, right? We've put ourselves into a position where it's like, you know, if

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the gut were a muscle, then the collective muscle of the American gut is so weak right now,

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right? You know, um I've been saying this for a long time, too. You know, a

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lot of the big data um points at these these kinds of trends. So, for you know,

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somebody that wants to pay attention to this and wants to incorporate more fiber

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in in their diet, not all fiber is created equally. you're a big fan of plant sources. Um, what are some of the

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go-to plant sources, let's say, aside aside from fermented vegetables? What are some of the go-to plant sources that

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um we could add to our diet to make sure that we're getting the right amount of soluble insoluble fiber?

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So, I think that the key here is that so f first of all, fiber can be incredibly complicated.

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I agree with you. Um, it is biochemical. People think it's all the same thing. No, no. It's like the word protein,

38:49

right? The protein in a fish is not the same as the protein in a bean. Right. Right? And the word protein is a

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big word. There's a lot of different types. There's a lot of different types of fiber. And but yet that complexity can

39:02

be distilled down to simple rules that everyone can take away from this podcast. Okay?

39:08

All plants have fiber. Fruits, vegetables, whole grains, seeds, nuts, and legumes.

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By the way, mushrooms also have fiber. Technically, they're not plants. They're fungi. But like, let's make them honorary

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plants. Yeah. Yeah. Let's make a moderate plan. So all that has fiber, okay? They also

Plant Sources of Fiber

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all have both soluble and insoluble fiber. When we use these words soluble and insoluble, it's just describing a

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characteristic of the fiber, right? And like another word that we could use is viscosity, which is how

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thick it is, like how gelatinous it becomes. Or another word we could use is fermentable. How fermentable is it?

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These are just different ways to describe properties of fiber, right? Um, now soluble fiber and

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insoluble fiber we do need to separate out a little bit because the way they function is different,

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right? Insoluble fiber is the grit. That's the roughage. That's the traditional view of fiber.

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Like I grew up, I'm sure you did too, where the way that I saw fiber was my grandma stirring an orange drink so that

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she could have a dump. Yeah, exactly. Right. Metamucil.

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Yeah. She was having her Metamucil so that she could have a poop. So true. We knew what grandma was up to there. Yeah. Yeah. So now that's that's the

40:17

grit. Definitely grow up with that grandma. Yeah. And that's the grit and that just runs right through us. That's the new science of fiber that I

40:23

want to talk about which is the soluble fiber, right? The new science that's showing us that soluble fiber actually dissolves in

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our drink. It it goes down into our intestines passes through about 20 feet of our intestines.

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We don't have the enzymes to break down soluble fiber, but our microbes do.

40:42

Okay? So it passes through our small intestine, arrives into the colon, completely intact, and then they get into a feeding frenzy because this is

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their preferred food. So they get to eat. Wow. Right. And then they like go to work in teams unpacking that fiber and releasing

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the short- chain fatty acids. All right. Now, you asked me like what what are the foods that I recommend? Let

41:03

me say like I I'm I'm more than happy to share some of my favorites. Like I love avocados, right? I've already had

41:08

avocados twice today. I thought you said tacos. I was like, okay. No, no, but um but I think that

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the actually the key takeaway is this. I want to I want to actually reshape the way that people think about fiber to

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stop thinking about it in grams. Okay? Right? We have been told like even me I mean during the show I've said we need

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to get this many grams of fiber. Mhm. Okay. Actually when it comes to our microbiome the key is variety

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because different types of plants have different forms of fiber. So yes, like beans and lentils, they are

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a wonderful source of fiber and resistant starches. Yeah. And polyphenols.

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So like that to me, those are concentrated sources. The reason why a lot of people struggle with them is because their gut microbiome is not

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ready to handle the amount of beans they just ate. All right. But um but if we if we were

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to take a step back, the American Gut Project was the largest study to date that allows us to draw connections

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between our gut microbiome and our our lifestyle pattern, our diet.

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And in that in that study, it wasn't whether you called yourself vegan or keto or any of these different labels

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that we apply to a person. That didn't matter, right? What mattered was that the people with

42:27

the healthiest guts were the people who were eating at least 30 plants per week. Wow.

42:33

And um that sounds like that's more than 30 servings, right? You're saying 30 plants, 30 total plants, 30 plant varieties.

42:40

Okay. So, and and the key with that is like no matter who you are, if you're not

42:45

hitting 30, don't feel bad because 95% of America is not doing this. Right. Right. So, start where you are. But just

42:53

think about this. When you go to the supermarket, when you go when you step into the kitchen and when you sit down at the dinner table, there's an

42:59

opportunity where you can add more variety. Right? You're making tomato sauce. You could do tomato sauce and pasta. That's two plants. Or you could

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take that tomato sauce and turn it into a vehicle for throwing all kinds of different varieties of plants in there. Yeah. And and when you start putting a

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wide variety of plants in there and wide variety of soluble versus insoluble fiber, you start producing these um

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short chain fatty acids. you see what kind of markers could we watch for in

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because this is the other thing is how do I measure whether or not I'm doing a good job with my microbiome, right? I

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mean, we're both big data guys and so I test my microbiome. I see that it's off.

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I make some adjustments to my diet. Um, what kind of biomarkers could I expect

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to to see in my blood and, you know, change? Well, so I guess let me let me sort of

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frame this, which is to say that I mean, yes, you could do a a microbiome test. I

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work with a company called Zoey, and we have an at home microbiome test that also includes a CGM. I've seen Zoe. Yep. So, our our our kit is combination

44:02

of microbiome, continuous glucose monitor, and a blood lipid test to basically create a personalized dietary

44:08

plan. Okay. Okay. And we have we have a randomized control trial that we published in May in Nature Medicine proving that our

44:14

product works. All right. But put that to the side for a moment and let's just start with something that's completely

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free and very basic, which is how do you feel? And what do your bowel movements look like?

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Right. This is where I would start because people that have a damaged microbiome, they have low energy. They

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have tons of fatigue. Yeah. They have difficulty with concentration and focus. Right. They can't get a lot done. They're

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struggling in their exercise and their workouts. I noticed this myself. I'm curious if you notice this. If I eat a

44:44

junk food meal, which like I'm I'm not perfect, right? I do the best that I can. If I have a bad meal, my workouts are way

44:51

worse on bedtime. Oh, yeah. No question. And if I have a bad meal, I just finished a podcast on this, but if I have a bad meal too close

Gut Microbiome Tests and Checking Yourself

44:57

to bedtime, it's even worse. 100%. You know, and not only destroys the mood, but also destroys sleep and then

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you pay for it the next day, right? Yeah. So, all of these things are the product of your microbiome. If you have gut

45:09

symptoms when you eat regular amounts of food, right? That's because your gut microbiome is struggling to help you to

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process and digest your food. Look at your bowel movements. So, you get you get byproducts that you

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don't want. Gas, bloating, constipation, irritability, cramping, all of these things that

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sloppy digestion, right? That we think are food allergies or food sensitivities. And then I guess in a sense because we don't have the bug

45:32

gut microbiome, they sort of are. Um but the the exciting thing is that you're

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not stuck with the microbiome that you have today. No. And there's common sense there's a common sense approach that you can use

45:44

to um address this issue so that people who want to consume more fiber

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but struggle to consume even normal amounts of fiber like the numbers that we're talking about they can accomplish this. And the way

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that I go about this is to use an exercise analogy. So like if you hurt

46:02

your knee, you have two choices. You could go lay on a bed

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not move. All right. And I promise you, you will not feel pain in your knee. Right?

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But your health throughout your entire body will suffer because you are completely sedentary. You're not using

46:21

the rest of your body at all. Eventually, this will manifest with disease and it will actually create problems for

46:27

you. Right? The alternative choice is that and I know Gary I'm sure when I ask this

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question it's a little more complicated with you because you got some other things you would do. Yeah. Right. But like for the average person

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the other choice is to go through a rehabilitation program. Right. Right. And in a re rehabilitation

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program you're actually working the knee. You're actually forcing it to move.

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It's controlled. Now you may have some discomfort when you do this but like that's to be expected. Right.

46:56

Right. and you're working through a process of restoring function to that knee so that by the time that you're done you can get back to playing pickle

47:03

ball or playing basketball or whatever it is that you like. Right? The gut is the same. So the problem with

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the the conventional approach that we have these days is that people say if you don't feel well then just eliminate

47:16

it. Oh yeah. And people end up on a progressively more restricted diet.

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They're shaving away more and more different types of foods. And I just said 10 minutes ago that the number one

47:28

predictor of a healthy gut microbiome is the variety. Abundance is the key, not restriction.

47:34

Right. Right. So, how do we go about this? The way that we do this is and we rehabilitate the same way you would with

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a knee, which is that you got to take these foods that you're struggling with and put it in an amount that you can

47:46

actually tolerate. So, start low and go slow. Mhm.

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Reduce the amount that you consume at one meal that causes the reaction. That causes the reaction.

47:57

Yeah. Make it an amount that you actually can tolerate. There is an amount that you can actually tolerate. All right. Make

48:02

it an amount that you can actually tolerate, your gut will grow stronger. Yeah. Because it grows stronger, you can

48:09

increase the amount that you can eat next time by a little bit. Yeah. It's just like exercise. Yeah. So, and when we say we're going to

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go slow, so you know, back to the um the biomarkers that that test because again, I'm a huge huge fan of data and I like

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to to measure and track things. You know, one of the things we do in our our clinic environment is we'll we'll we'll

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pull a blood panel um and pull a genetic profile and then we will supplement for

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the genetic deficiencies and then watch the changes in the blood. So we're looking at things like C reactive protein homocyine

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um kidney eGFR you know filtration rates um alkaline phosphotase in the liver you

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know the the enzymes in the liver as alt seeing if they're the liver is getting more or less inflamed so we and in large

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pools of patients you can see that things like well when I take glutathione

49:04

and I support my healthy detox pathways we actually see a reduction in alkaline phosphotase and usually

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reduction in enzyme activity that looks that that indicates inflammation and

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irritation. So if I'm if I get a gut microbiome test and I want to go on a

49:21

journey to fix my gut microbiome or I'm listening to this podcast and I want to start incorporating things in my daily

49:26

life that are actually going to have an impact. Uh what sort of things should I be measuring? Um and I'm going to put

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some of these tests that you suggest. I'll put them in the show notes. Um because I think there's so much out there on gut microbiome that's hard for

49:38

people to to figure out where they start. Well, I think that the the reason why it's confusing is that the gut

49:45

microbiome um it has functional abilities and those functional abilities apply to all different aspects of who we

49:52

are in our health, right? So, it's not just this one measure. How is your gut microbiome doing for your digestion, for your

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metabolism, for your immune system? Like these are separate separate goals and we

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need a microbiome that supports us in each of these goals, right? So where you have to start is you have to start by first saying what do you

50:09

what are you hoping to accomplish? What what do you want to go after? Right? Do you want to enhance your metabolism? Do you want to enhance your

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immune system? Do you want do you want to be better at digestion? Right? Starting with a holistic before

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we get into into blood measures just real quick and and I kind of touched on this but just to be totally clear I

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start with how does a person feel? Right? What is their health history? Do they have what are their symptoms?

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What are their symptoms? What is their health history? Do they have a history of diagnoses that are associated with uh dispiosis? Because

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like when I would go to see patients, I would take a look at their health history and you see like they're here to see me for irritable bowel syndrome, but

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they also have major depression, ADHD, right? They're they have obesity, they have high cholesterol, they have

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pre-diabetes, um they have polycystic ovary syndrome. Okay, everything is out of whack,

50:56

right? But yet the one thing that ties all that together is the microbiome. Phenomenal. Right. So, and then the last thing is

51:03

like what's going on with your bowel movements? Because if I were a cardiologist, I would listen to your heart, but I'm a gastronurologist. And

51:09

your poop is predicated on a healthy microbiome, right? 60% of the weight of your bowel movement

51:16

comes from microbes. Mean the byproduct from microbes. The the waste from microbes.

51:21

The waste from microbes. Yes. So, we think of our poop as being the excrement of our food. It's our food waste.

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But actually, the majority of the weight of our food is microbial in nature. And one of the ways that you know, and I

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can prove this to you because if I if I were to put some soluble fiber into

51:38

a drink for you, you would have a vigorous large bowel movement. Not not while we're live and and on the

51:44

air here. Vigorous large bowel movement. Yeah. I I feel quite comfortable talking about these things. Gary,

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you just let that fly, dude. I love it. Oh, I I I I bet you there's some people watching

51:56

this right now. They're like, "Hey, if I can have a vigorous large bowel movement, I am in. Tell me where I can

52:01

get that because constipation is a massive issue with I can't even tell you

52:07

how many clients come into our clinic." Oh, I'm convinced it's epidemic. Epidemicize

52:14

one. Two. I mean, and there are people who poop every day. I'm every morning like I'm like a bear.

52:20

No, but there are people who poop every day and they're still constipated. Oh, okay. Yeah. So, it's not it's not even just your your frequency like there's

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So, but I think that the key here though is that a healthy microbiome is reflected in good healthy large

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satisfying bowel movements. Yes. All right. So, we start with those things.

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All right. And then the question is okay. So, which which index which which direction do we want to look into?

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Because if what we want to look into is your digestive health, then to me as a gastronologist, it's about symptoms.

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That's what I treat as a gastronologist. I treat symptoms. That's the focus, right? But if we were looking at your metabolic health and we want to say where's your

52:57

metabolism at, then I want to take a panel of labs that are going to show me where you're at in terms of your metabolic health.

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And if if I have to start somewhere, I want to look at insulin sensitivity. Insulin sensitivity. Huge. That's where I want to start. I mean,

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but we could also, of course, take a look at at your lipids, take a look at your cholesterol. We could get a full

53:14

panel. Mhm. Um, one of the things that I that I like with our Zoey program is that our um,

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our triglyceride test is a postprandial triglyceride test after a meal.

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So, it's after a meal. It's a standardized meal. So, that way everyone who does the Zoey kit, whether

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it's you or me or someone else, they get the same meal and then we can see what the response

53:36

is. Now, this is like to me conceptually goes through the microbiome. What shows up in the blood

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is what you're you're measuring that response. So, believe it or not, our our our blood fat uh levels, this is

53:48

something that we published in 2020 in Nature Medicine, our blood sugar, our blood fat,

53:54

specifically triglycerides, and our insulin levels can all be predicted by our gut microbiome.

54:01

Wow. On different levels. On different levels. So, um the most powerful thing for blood sugar, for example, is what

54:08

did you just eat? Right. So, if you eat like a high sugar meal, but that's just glucose, right? Right. But hemoglobin A1C has sort of

54:15

what you're eating over times and you know much longer window 90 days or you know three-month window of your blood

54:21

sugar. That's right. Um but you're saying that's highly predicted by your gut microbiome. So I'm saying that your blood sugar is highly predicted by your gut microbiome.

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I'm saying that even more so your insulin and the reason why insulin is important is because actually the amount

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of insulin that you require to process the exact same food is directly indicative of your insulin resistance or

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insulin sensitivity. Wow. Right. So, and that's that's very much predicted by your gut microbiome. And

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this aligns with the research that shows that if you take the microbiome of a

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person with diabetes and you transfer to a mouse, that mouse will get diabetes. And yes, Gary, even in humans, we have

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data where they took young men, gave them a fresh microbiome. They had pre-diabetes, gave them a fresh

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microbiome, and they they saw improvements in their in their blood sugar. Uh and when you say gave a fecal transplant

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or or or probiotics, prebiotics, what did they do to give them a microbiome?

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So in in that in that particular study where they gave it to the young men, it was a fecal transplant. Okay. But if I were given the choice, what

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would I rather give a person? A fecal transplant or or a prebiotic. I would rather give them the prebiotic,

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right? And the reason why is because the fecal transplant um so let's pretend in theory like if if

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I were to give you a fecal transplant, what would happen? Well, the issue is that the new microbes that I've given

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you are not uh aligned with the food that you eat, right?

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They're aligned with the food of the host of the host. Right. Right. So, you may derive a benefit

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initially from having this fresh microbiome that maybe has functional abilities that that you didn't

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previously have. And I don't mean to pick on you, Gary. It's just like in theory. Yeah. Right. So, you may derive a benefit from

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this. So like in this study for example, these young men who had a blood sugar problem, they got a new microbiome and

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suddenly their blood sugar problem got a lot better. That's phenomenal, right? But but the issue though is that they never changed their diet

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and because they never changed their diet over the course of four weeks, they shifted back to their old

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microbiome and they were right back to where they started, right? So it's not a sustained benefit.

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So if you wanted to sustain the b the benefit, what you would need is you would need the prebiotic. you would need

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to shift your diet. And by feeding the microbiome, then you would actually be able to maintain that benefit and

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perhaps even go beyond it. Hey guys, I'm really excited to announce this. Perfect aminos has gotten a

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serious upgrade. They've added nucleotides, the building blocks of our nucleic acids, DNA and RNA. And this is

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important. We know essential amino acids are the building blocks of protein and collagen. Having all the essential amino

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acids in the correct ratio is necessary for complete protein synthesis without

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the caloric impact. But if we want perfect protein synthesis, we need to look at the process of protein synthesis

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itself. Because if the process is faulty, we won't get the correctly made protein, collagen, fibbrin, or the red

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blood cells in our bloodstream or our muscles. We can even stop creation of specific proteins which can affect us in

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so many different ways. Our DNA and our RNA are what direct protein synthesis,

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building new proteins. If our DNA or RNA get damaged from toxins, harmful

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bacteria, or just plain aging, we get faulty protein synthesis. So cells,

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enzymes, and hormones are less functional and we get premature aging. By adding nucleotides and nucleotides,

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the building blocks of the nucleic acids DNA and RNA, our cells get exactly what

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they need to help repair faulty DNA and RNA and improve the process of protein synthesis itself. This is next level

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science and you need to try these. Now, let's get back to the Ultimate Human podcast.

58:01

Wow. So, what are some like musthaves, some go-to foods um that you say, you

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know, basic healthy gut nutrition, basic healthy microbiome function is is based

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upon maintaining this level of fiber, this level of soluble fiber. Here's the best sources of soluble fiber in your

58:23

opinion. I mean, if I if I have to pick one, I'm going with legumes, right? And yeah, and the reason why so

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lentils, peas, and beans. Okay. And the reason that I'm going to go there is also part of a lot of the longevity

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diets. I mean, all all the longevity diets show this. Yep. Walter Longo's research supports that.

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There are research studies where they um control for multiple different factors and then they spit out like what is the

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number one food and and beans are the food that pops out? Um what is it about beans that's

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special? Well, um they have a tremendous amount of fiber and also resistant starches

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and they're being wildly under consumed. So the average American, you know, like I I I laugh when people propose the idea

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that the lectins and beans are the cause of all of our issues. Yeah. I mean, this makes zero sense.

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You do hear that for sure. Well, Gary, we the average American eats six pounds of beans per year.

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Yeah. Per year. Our parents our parents generation ate eight pounds

Basic and Healthy Gut Nutrition

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of beans. We've reduced our bean intake substantially. Mhm. This is not the cause of our problems.

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Okay. Okay. So, but it is an opportunity. It is an opportunity because these are foods that are non-p perishable.

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Completely inexpensive, right? Like there's very few foods that are less expensive than beans.

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Yeah. Yeah. Legumes are really cheap and they can be worked. They're pretty versatile. Yeah. They're very versatile. So, they can be

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worked into our diet in a number of different ways. It doesn't need to be this thing where you're on an all bean diet. That's not what I'm proposing,

59:55

right? What I'm proposing is that you actually reintroduce these into your diet and just have them in there. Right.

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Right. In addition to everything else that you're eating. So, legumes, um, you know, fermented,

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um, veggies, which is one of my favorites. I I I remember I hated them so much as a kid. I crave them now as an

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adult. They are they they go with just about anything. And sauerkraut can be really really powerful, but some of the

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kimchi are delicious and they're crunchy and like, you know, if you have a hot meal on your plate, you can have them

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cold. You can actually make them hot, too. Yeah. Um and so these kinds of things, but um

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is soluble and insoluble fiber because I think when we talk about gut microbiome

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health, most people think they go straight to the shelf and grab a probiotic. Um and the way that you know

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most people would evaluate probiotics is they go, "Okay, how many col forming units are here?" Oh, 50 billion CFUs.

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That's good. And um it has this one has seven strains and this one has 11 strains. So more colony forming units,

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more strains, and it's more expensive. So it's probably the best probiotic. So

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kind of walk us through that because what's interesting about the probiotic research that I've looked at is it seems

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to be almost black and white, meaning there are those these probiotic studies where had no impact um on the

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hypothetical outcome and then there were probiotic studies where it had a demonstrative impact, some of which were

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fecal transplant studies. And I and and to me that meant it must have something to do with how the probiotics were

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getting in. Maybe they were getting destroyed by the stomach acid. Um maybe they weren't the cultures that they

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reported to have in in the probiotics. So kind of walk us through that maze because the probiotic section of the

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most nutritional stores right now is, you know, it's like going into a 7-Eleven trying to find an energy drink.

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There's a lot of choices. Yeah. Right. Um so there are definitely people who benefit from probiotics. I've taken care

Probiotic Studies

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of these patients. I've had many patients that have recommended probiotics too and seen them have amazing results.

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All right. But um it is not one sizefits-all and you will not know until you actually

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try the probiotic and see what it does for you. Okay. So, and the reason why this is is

1:02:04

because Gary, you have a you have a gut microbiome that's completely unique to you.

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There's no one on the planet. This is as as much of a signature as your

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fingerprint. M and if you had an identical twin in theory, like if you had an identical twin, you would only share about 30 or

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35% of the same microbes. Wow. You would be more different than you are the same, which could explain some of the

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differences between identical twins that are genetically 100% identical. Right. 100%. So yes, they're genetically

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identical and yet they have like there can be wild discordance in terms of many different health measures, obesity or

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other things from health perspective. This is this is actually how um someone that I work with, Tim Spectre, he got

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started in the space because he was looking and doing twin studies in the '9s in the 2000s and what he discovered is like I can't

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explain it all with genetics. There's something else going on here. Yes. And so and then that's what led him into

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the microbiome. Microbiome is kind of epigenetics, right? I mean it's it's going to have an influence our genes predisposition but I

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mean the microbiome this is an important point because I actually didn't mention this earlier which is that this is an an important point which is that um we have a genetic

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code that genetic code is not the story of our life and it is not whether or not we

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will or will not develop disease right um it is a switchboard and those switches can be turned on or

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turned off which is epigenetics is what you're talking about and the question is who's sitting at the

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switchboard That's powerful, right? And the answer is the microbiome. Wow. The microbiome has the ability through

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short- chain fatty acids to actually modulate our genetic code and turn certain genes on or off.

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That's so fascinating. I mean, what what is, you know, in this in this field of the microbiome and gastronurology? Like

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what what is inspiring you right now? Like what's really captivated your intention? What roads do you really

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really go down? Um what do you think the the the population needs to know because

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I mean you're you know this is this is on the fringes of mainstream modern medicine for sure. Yeah. Right. I mean in terms of mainstream

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adoption of modern medicine. Sorry. Yeah. I mean I think I think that we you know I'm excited by these amunotherapy

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studies that we're talking about. Right. I like I think vaccines imunotherapy. I'm

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fascinated by this. I also pray though that people don't need this. Right. I also pray that people don't actually develop disease

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that we could prevent it before it ever happens. Right? So to me, what's exciting about the microbiome is not what we do after a

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person has cancer. Although I although I want that to be successful, to me what's exciting is the opportunity

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that we have through common sense choices, simple choices where we could add more varieties of

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plants to our plate, where we could add fermented food back into our diet, right? where we could get into a routine

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of regular exercise, where we could align our circadian rhythm, where we could get a good night's rest and

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suddenly nurture a healthier gut microbiome that prevents us from getting the melanoma in the first place. Like,

Preventative Measures

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that to me is the great opportunity that we have. Yeah, I I I love that, too. Um, I feel

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like we have a whole second podcast brewing here, which I'd love to invite you back to to to do. Um, we're we're

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we're going to take a little detour and jump into what I call our rule breas and and answer some questions. But, um, for

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my audience that's really interested in your topic, where can they find out more about you? Where can they where can they find some of the supplements that you've

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um come up with your book? How do they find you? Yeah. So, um, you can follow me on

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social media. So, I am the gut health MD on on Instagram, Facebook. I'm on TikTok

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uh as the gut health MD because some kid took my handle before me. So, stole it from you

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and and if you're out there and he's the gut health MD. Well, they're not they're not posting anything. I don't know what they're expecting to happen, but

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yeah, you'll get a message that they're holding you hostage. For five grand, you can get it back. Yeah. Exactly. Exactly. Um so, you can find me

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on social media. Um I'm very proud. My favorite thing, honestly, Gary, is writing emails.

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So, because I feel like writing is the ultimate form of communication. Mhm. PE most people who follow me follow me

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because of my books. My first book, Fiberfueled, I don't even know how many sold. It's like certainly more than

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300,000 copies. And um so and I spent a year writing that book

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and you could sit down in a weekend and like hang out with me and learn, right? Yeah. That to me is like what writing is

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about. So I enjoy I enjoy writing emails to my community. I have a community of about 200,000

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people on my list. Awesome. and we, you know, basically like dig into these kind of nerdy topics that I love and we talk about poop and

Connect with Dr. Will

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all this different stuff, right? So, I would encourage people, it's a completely free resource.

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You owe me nothing, right? That you can come and sign up for my email list and learn with me.

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Yeah. Right. With this community, give without the value. Uh, you know, my my belief system is

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that I I think that, you know, I believe in a food first approach. Like, I think that nutrition needs to come first,

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right? I don't think that you can outsupplement a bad diet. I totally agree with that. Nor should you try.

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Um, but I do believe that when we when we do nutrition and when we uh engage,

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you know, with these lifestyle things like exercise and sleep, there are limits to how far it takes us.

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And oftentimes there there's also opportunities for more where we could optimize,

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right? Where we could fill in the holes that haven't been properly filled by our diet, which exist in all major dietary

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patterns, right? You and I could sit here and poke the holes in all of them. I tell people all the time like you have

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no idea how good normal feels till you really feel normal. I totally agree. It's amazing. So, I'm of the belief that there's a

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place for supplements. Supplements are not a replacement for food. Supplements are in addition to food. And I'm just speaking from my own

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personal experience. No one will ever be able to convince me otherwise of this that I sat there for 10 years and I

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watched people as I put them on fiber supplements, many of them who were already eating a healthy diet because

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this is what I became known for, right? So they sought me out and they were already on a healthy diet trying to

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get better and the fiber supplements made a radical had a radical effect in terms of their health. So, but the but

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the issue that I always saw, Gary, is that the supplements, like you know, that currently exists, I'm not going to name any names, but you you all know

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which ones I'm talking about. Um, they've existed since I was a kid. We're talking about outdated 1980s

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formulas, right? These are single I'm I'm saying that diversity of plants is the key. Yet, this is a single source of fiber

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that's being sold to us by a pharmaceutical company that's not third party testing it or telling us anything

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with transparency. Right? So to me, there was this opportunity to do so much better for

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people who want to take a prebiotic supplement. So that's why I decided to create my own and and I launched it in January of this

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year for the first time and it went so fast that we had to close the store after 24 hours. Really?

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Yeah. Because I was going to sell out of my stock and it takes me about six months to restock. So um but now here we are. We've

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restocked and um we the the the brand is called 38 terra 38 ter.

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38 terara stands for 38 trillion the number of microbes that live in our gut. And basically I spent three years

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formulating the exact prebiotic supplement that I would want for myself or for my patients

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because it gives them fiber resistant starch and polyphenols. And these three things are unique forms

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of prebiotics. They're not the same and they synergize with one another. So like basically the polyphenols, you

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and I were talking about this beforehand. Yeah. The polyphenols shape the microbiome to actually increase their ability to

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create short- chain fatty acids from the fiber and the resistant starch. Wow. So um and it's been very satisfying to

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do this because um the way that I see my career is that I'm trying to create

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change on mass scale. Same. Absolutely. And and change comes in the form of a person reading my book

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and changing their diet or listening to this podcast and taking away some actionable tips

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or adding this supplement and then writing a review that's a fivestar review and telling me how they cleared

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up their psoriasis or they improved their gut issues or they've never been better in terms of their inflammatory balance. To me, this is what it's about.

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No better feeling in the world. Well, I end every podcast by asking every guest the same question. There's no right or

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wrong answer to this question, but what does it mean to you to be an ultimate human?

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I'm gonna go in a totally different direction than we've been, right? I mean, we could go ahead. We could we could talk about the

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ultimate human being a human that embraces the whole person. That includes our microbiome, right? We are

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um uh super organism. Mhm. But actually I want to talk get back to

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um whole uh the ultimate human is a person who's connected to others because that

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actually is the source of our happiness. We are social creatures

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and longevity. If you want it's as powerful as what we eat. Yeah. Right. And if you want to um torture a

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person you isolate them. Amen. And so, and I think in today's world, I see a lot of sickness, a lot of

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despair, and a lot of people who are sad. And I think that part of the way that we heal is to put down the phone

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and look a person in the eye, connect to that person, right? And feel loved,

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right? That's what we need and feel connected and appreciated. And yeah, that's fantastic. That's one of

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the best, if not the best explanation I've gotten so far. Um, it's good for our microbiome people. Yeah. Of course, you had to bring it

Final Question: What does it mean to you to be an “Ultimate Human?”

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back to poop, but we can always bring it back. Thank you so much for coming on, man. I'm I'm definitely going to invite you

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back. I I think that your journey's just begun. Um, and I think that people like you that have had so many profound

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experiences in their life and they're actually turning that into tangible ways to touch more lives. Um, you're the kind

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of person I really enjoy giving a voice to and I'm going to follow your journey and bring you back in front of my

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audience, you know, again, very soon. Appreciate you, Gary. Appreciate you too, Ben. And as always

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guys, that's just science.